

Department of Computer Science
Gujarat University
M.Sc.(Artificial Intelligence & Machine Learning) – III

Fasal Rakshak

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Introduction

During automated agricultural harvesting, animals hiding within dense crops often suffer accidental injury or death due to limited driver visibility and poor field contrast.

To address this critical issue, this project develops a real-time safety monitoring system powered by camera vision and deep learning object detection. By analyzing live video feeds from the field, the system automatically detects hidden animals and hazards in real time. Visual alerts and detection overlays are rendered directly onto an operator user interface, enabling machinery drivers to pause operations immediately and protect field wildlife.

Harvest Mortality in Numbers

~500,000

wild animals killed per year by farm mowers in Germany (estimate), incl. ~90,000 roe deer fawns [1]

25–44%

of roe deer fawn mortality attributed to fodder-crop mowing in a Swedish study [2]

2.6 million

birds killed per winter by olive-harvest tractors in Andalusia, Spain (press report) [5]

Hungary mowing survey (carcass shares) [4]
Reptiles 57% · Mammals 30% · Birds 6%
Central Europe: tens to hundreds of thousands of fawns per year [3]

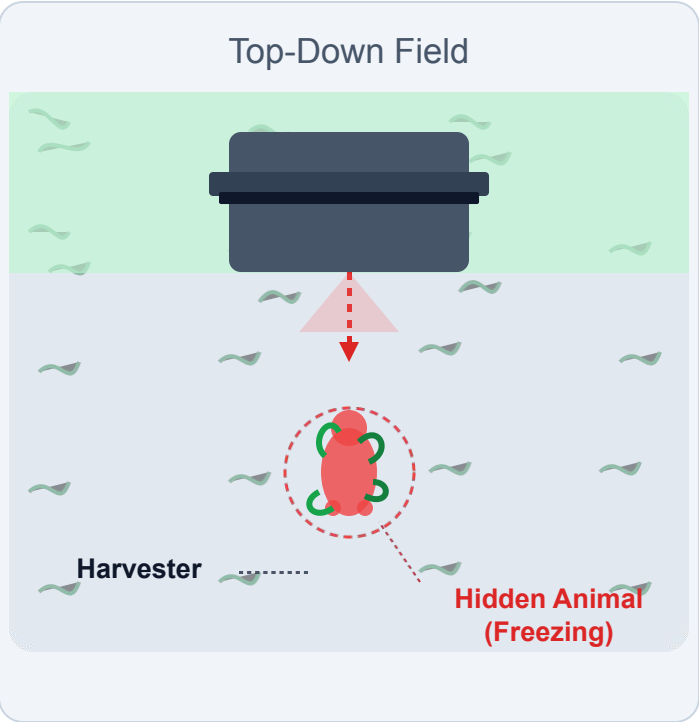
India and Gujarat
No species-wise harvest death count found in the literature reviewed.
Only indirect signals: 110+ leopard cubs reunited after cane harvest (NGO) [6].

No worldwide total exists; figures are regional estimates from Europe. Several are NGO or press based, see references.

1. Harvest mortality of fawns, nilgai, and ground-nesting birds

2. Animals freeze instead of fleeing, so operators and RGB cameras miss them.

3. **Idea:** sense what humans and RGB miss, using **IR-based vision with AI**



Research Problem

Problem Statement

We need to find animals hiding in tall crops before a moving harvester runs them over, using a camera mounted on the front of the vehicle.

Blind in Thick Crops: Existing cameras and AI fail to detect animals when they are heavily covered by thick plants (like Indian sugarcane or wheat).

Limited AI Training: Most AI models only know how to find one type of animal, and there is no public data for hidden wildlife in Indian farms.

Too Expensive: The systems that *do* work well cost way too much for regular farmers to afford.

Target Animal Classes

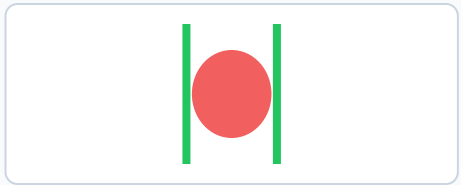
Class	Why chosen	Evidence
Nilgai	Major crop raider in UP and Uttarakhand; large, warm-bodied	Crop-raid surveys [7][8][9]
Wild boar	Shelters in tall crops; persistent raider; large and warm	Bundelkhand raids [9]; 309 human casualty cases, 5 states, 1990–2008 [10]
Snake	Common in Indian fields; reptiles were the largest mowing-carcass group elsewhere	57% of carcasses in Hungary [4]; near-ambient body temperature (hard thermal case)
Stray cattle / calves	Abundant in UP fields; warm-bodied; data-feasible	Stray cattle crop damage [9][11]
Small mammals (hare, jackal, jungle cat)	Recorded in Indian cane landscapes; heavily hit by mowing abroad	Maharashtra cane study [12]; 30% of Hungary carcasses [4]
Big cat (leopard), optional	Uses cane as shelter; safety-critical	Cubs found at Dec–Mar cane harvest [6]; rusty-spotted cat kitten [13]

Honest note: no study found counts harvest deaths by species in India or Gujarat, so classes were chosen for presence, risk and detectability.

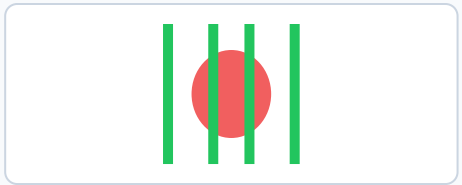
RESEARCH QUESTION

How can single-stage object detectors (e.g. YOLO) be structurally modified to accurately localize fragmented, occluded thermal signatures from a ground-level harvester vantage point, and where, as a function of crop density, does that detection reliability break down?

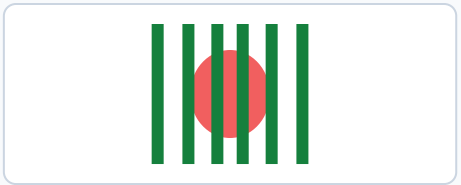
Progressive Occlusion by Crop Density (Ground-Level View)



Low Density: Minimal Occlusion



Medium Density: Partial Occlusion



High Density: Severe Fragmented Signature

Literature Review & Research Gap

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Approach	View	Occlusion	Limitation
Aerial	Top-down	Canopy-level	Below 25% detection under dense canopy
Perimeter	Static boundary	Outer edge only	Misses internal field activity
Roadside	Angled static	Line-of-sight	Limited range & static coverage
Lightweight IR	Thermal sensor	Partial thermal	Requires edge optimization for motion
Harvester	**Ground-level**	**Progressive occlusion**	**Research Gap: Forward-motion detection**

Research Gap

- **No harvester-mounted, in-crop detection system:** All reviewed systems operate either aerially (pre-harvest scouting) or from vehicles/farm perimeters - none detect animals already hidden inside standing crop at the moment of harvesting.
- **Limited dense-vegetation/canopy testing:** Multiple studies (Winsen et al., 2022; Rietz et al., 2023) directly show detection probability collapses under dense canopy/occlusion - a core unresolved challenge for high-density crops like sugarcane.
- **Predominantly single-species/animal-class models:** Most detectors (Lyu et al.; Puppala et al.) are trained for one species, limiting applicability to the variety of animals that hide in crops.
- **Scarce public, harvest-specific thermal datasets:** No dataset in this review specifically captures partially-occluded animals inside standing agricultural crop; most groups rely on custom, largely unreleased data.

- **Metric–reality gap in real-time claims:** Povlsen et al. (2024) explicitly show high mAP does not guarantee correct real-world detections, indicating a need for field validation beyond benchmark metrics.
- **Limited thermal-sensor resolution/range on affordable hardware:** Puppala et al. (2025) report reduced detection range (~100 ft) and accuracy loss in high ambient heat with low-cost thermal sensors.
- **No Indian agricultural crop-specific datasets:** Farm-intrusion studies conducted in India (Sathesh et al.; Goyal et al.) focus on perimeter intrusion, not in-crop detection, and no dataset specific to Indian crop types (wheat, sugarcane, paddy) with hidden-animal thermal signatures was found.
- **High hardware cost for research-grade thermal/UAV systems:** Professional-grade UAV thermal payloads used in several studies (Lyu et al.; Rietz et al.) represent a significant cost barrier for widespread farm-level adoption.



Objectives & Scope

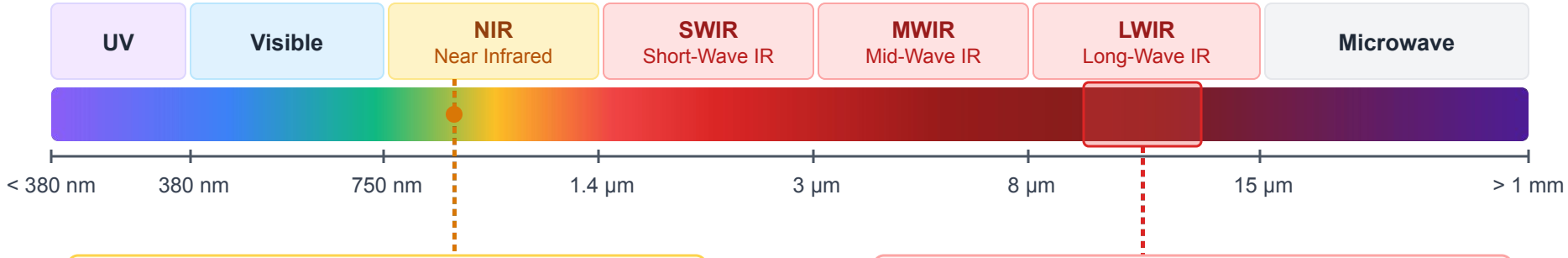
1. Design a deep-learning-based thermal detection pipeline capable of identifying animals partially occluded within a standing crop.
2. Adapt real-time, edge-deployable object detection architectures (e.g., YOLO-family models) proven effective in UAV/vehicle thermal detection literature to a ground-level, harvester-mounted viewing geometry.
3. Evaluate detection performance across varying crop densities (low, medium, high) to characterize the operational envelope of thermal detection under occlusion.
4. Generate real-time operator alerts to prevent harvester-animal collisions.

EM Radiation, Wavelength and IR Bands

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Electromagnetic Spectrum — IR Regions for Wildlife Detection

Wavelength categorization highlighting active NIR illumination vs. passive thermal LWIR sensing



850 nm Marker

850 nm - IR illumination used by our CP-E25A camera

- Active near-infrared light source
- Requires camera sensor without IR cut filter
- Invisible to human & most animal eyes
- Does NOT detect thermal heat signatures

8–14 µm Region

8–14 µm - common LWIR thermal imaging band

- Passive thermal radiation emitted by warm bodies
- Operates without any external illuminator
- Detects heat signatures in total darkness / fog
- Requires specialized microbolometer sensors

KEY DISTINCTION: 850 nm IR illumination ≠ thermal imaging

CP-E25A uses active 850 nm NIR reflection, NOT passive LWIR thermal radiation.

Technology	Wavelength	What it sees	Performance for Hidden Animals
Visible (RGB)	400–700 nm	Standard visual light (human eye)	**Poor** -Blocked by dense crops
Near Infrared (NIR)	700–1000 nm	Vegetation health & moisture differences	**Limited** - Minimal foliage penetration
Short-Wave Infrared (SWIR)	1.0–2.5 µm	Plant stress, moisture & atmospheric haze	**Moderate** - Better than RGB; non-thermal
Mid-Wave Infrared (MWIR)	3.0–5.0 µm	Direct heat emitted by warm objects	**Very Good** - Good For low and midum dense crops
Long-Wave Infrared (LWIR)	8.0–14.0 µm	Radiated body heat / thermal signatures	**Optimal Choice** - Good For low ,medium and high dense crops



RGB vs NIR vs Thermal: Sensing

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Imaging Modalities — RGB vs NIR vs Thermal

Comparison of optical wavelength bands, sensing mechanisms, and operational strengths for agricultural wildlife monitoring

1. RGB — Visible Light



- **Wavelength:** 400–700 nm
- **Detects:** Reflected visible light
- **Output:** 3-channel RGB color image
- **Strengths:** High clarity & detail in daylight
- **Limits:** Fails in dark or visual camouflage

2. NIR — IR Night Vision



CP-E25A: 850 nm IR LEDs + CMOS Sensor

- **Wavelength:** 850 nm (Active NIR)
- **Detects:** Reflected IR illumination
- **Output:** Monochrome intensity image
- **Strengths:** Night vision via active IR
- **Limits:** Subject to physical occlusion

3. Thermal — LWIR



- **Wavelength:** LWIR ~8–14 µm
- **Detects:** Emitted infrared radiation
- **Output:** Radiometric intensity map
- **Strengths:** Highlights warm bodies in dark
- **Limits:** Lower spatial resolution / high cost

SENSING MECHANISM COMPARISON

RGB → Reflected Visible Light | **NIR** → Reflected IR Illumination | **Thermal** → Emitted Infrared Radiation



What Pixels Store?

What Does a Pixel Represent?

Technical breakdown of pixel representations across RGB, NIR, and Radiometric Thermal imaging

1. RGB Image



5×5 Pixel Patch

(128,76,52)	(130,80,55)	(125,72,50)
(210,185,140)	(205,180,135)	(198,175,130)
(115,160,85)	(118,165,88)	(110,155,80)
(95,140,65)	(100,145,70)	(90,135,60)
(140,90,60)	(145,95,65)	(138,88,58)

RGB Pixel = 3 Intensity Values (R, G, B)

Core Concept:

R + G + B → visible-light color information

Key Characteristics:

- Wavelength: 400–700 nm
- 3 visual color channels per pixel
- High spatial detail in daylight
- Fails completely in darkness

2. NIR Image



5×5 Pixel Patch

82	85	117	120	125
140	165	203	210	195
220	241	235	215	180
90	95	105	110	100
60	65	70	75	68

NIR Pixel = 1 Intensity Value (0–255)

Core Concept:

Pixel intensity → detected reflected NIR signal

Key Characteristics:

- Illumination: Active 850 nm IR
- Single grayscale intensity channel
- Night vision without visible light
- Does NOT measure heat

3. Thermal Image



5×5 Patch (Counts)

7420	7435	7580	7600	7590
7810	8120	8450	8480	8200
8510	8620	8590	8410	8100
7500	7520	7610	7630	7550
7380	7390	7410	7420	7400

Thermal Pixel = Radiometric Signal

Core Concept:

Radiometric signal → calibrated temperature estimate

Key Characteristics:

- LWIR band: 8–14 µm (thermal flux)
- Raw digital counts / radiance data
- Conversion to °C requires emissivity
- Pipeline-dependent representation

SUMMARY COMPARISON

RGB: 3 channels (R, G, B color) | **NIR:** 1 intensity channel (reflected 850 nm) | **Thermal:** 1 radiometric channel (emitted heat flux)

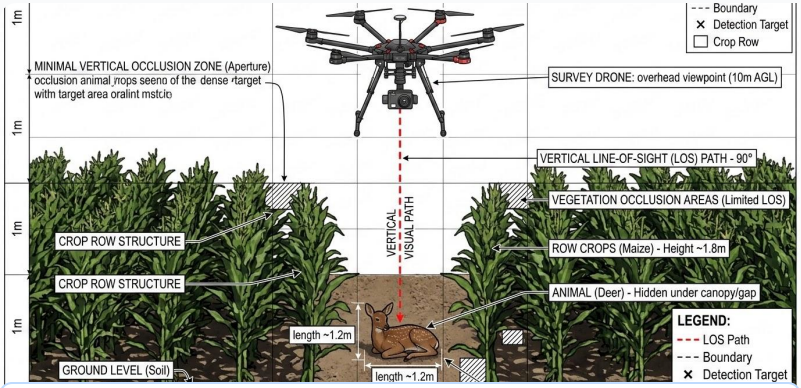


Why IR Helps Under Occulsion and Where It Fails?

Why Ground-Level Detection Is Difficult: Vegetative Occlusion

Comparative line-of-sight analysis between aerial drone surveillance and forward-facing harvester sensors

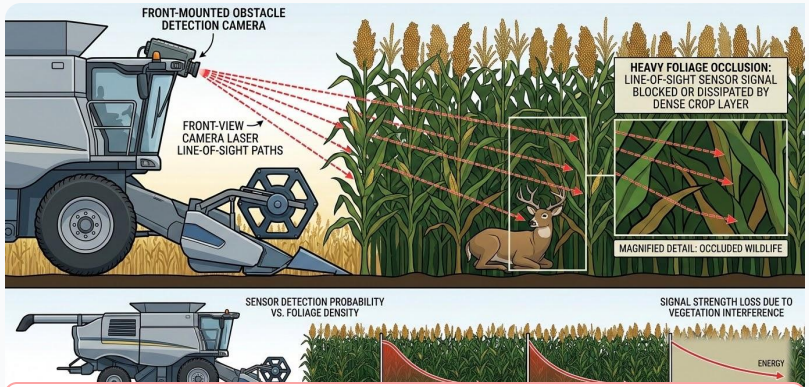
1. Aerial / Top-Down View (Drone)



Key Aerial Characteristics:

- **Vertical vegetation path** is comparatively shorter
- Direct top-down optical line-of-sight
- Minimal canopy overlap directly above target animal
- Continuous, unbroken IR heat & optical signature

2. Ground-Level / Forward-Facing View (Harvester)



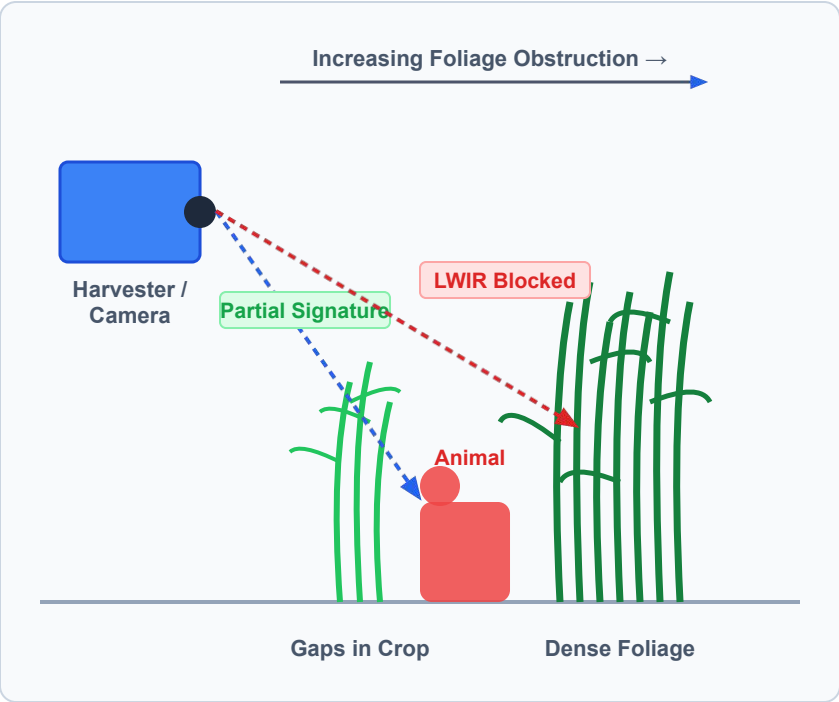
Key Ground-Level Callouts:

- **Longer horizontal viewing path** through dense vegetation
- **Severe occlusion:** Camera mounted at front faces dense crop wall
- **Fragmented visible IR signature:** Target partially hidden
- Physical barriers accumulate along horizontal sensing axis

CRITICAL SENSING TAKEAWAY

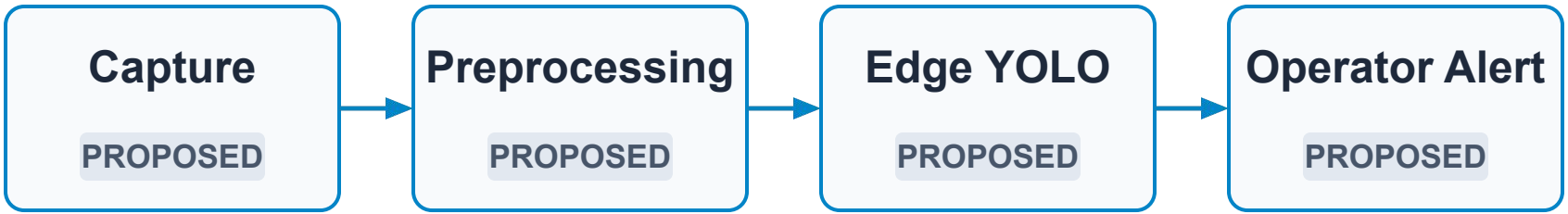
The same animal can be easier to observe from above than from a forward-facing harvester.

- Contrast comes from emissivity/temperature (thermal) or reflectance (NIR)
- Gaps in the crop expose partial signatures
- Foliage blocks LWIR, so results depend on density





Proposed System Architecture


Capture
PROPOSED
Preprocessing
PROPOSED
Edge YOLO
PROPOSED
Operator Alert
PROPOSED



Dataset & Evaluation Plan

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Why Synthetic Generation?

Key problems and justifications for simulated thermal dataset creation

1

Ground-Level Data Void

Existing datasets focus on top-down UAVs or static perimeters. Zero public datasets exist for forward-facing harvesters.

2

High Hardware Costs

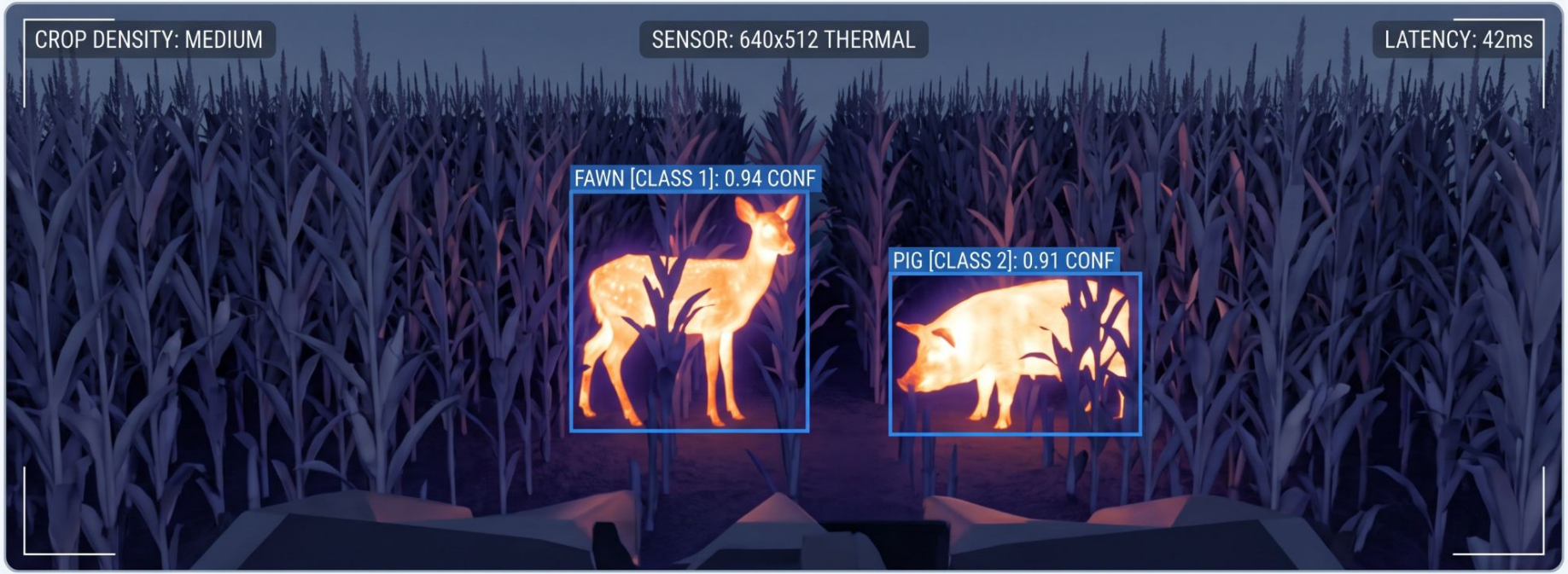
Industrial-grade Thermal IR (LWIR) cameras and specialized ruggedized harvester mounts are cost-prohibitive.

3

Precise Occlusion Benchmarking

Simulators allow full programmatic control over light, medium, and heavy crop density to test detection failure limits.

IDEA 1: BLENDER-BASED SYNTHETIC THERMAL DATASET GENERATION



TECHNICAL PROCESS FLOW (IDEA 1)

3D ASSET IMPORT



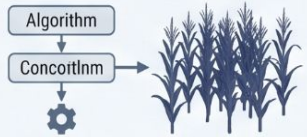
Import 3D Models

THERMAL SHADER & SENSOR SIM



Apply Emission (Heat) Shader
Simulate Sensor Resolution & Noise

PROCEDURAL OCCLUSION

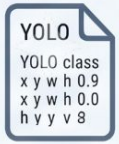


Control Crop Density
(Light, Med, Heavy)

SYNTHETIC OUTPUTS (Labeled)



Labled Thermal Frame



YOLO Training Image +
Automatic Label Export

IDEA 2: LAYERED THERMAL COMPOSITING SYNTHETIC DATASET GENERATION



TECHNICAL PROCESS FLOW

STEP 1: THERMAL VIDEO CAPTURE (RAW FOOTAGE)



Gather Field Footage (Thermal)

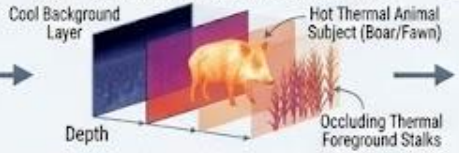
Gather Subject Video (Thermal)

STEP 2: THERMAL SEGMENTATION (LAYER ISOLATION)



Detect and Separate: Animal subjects, Foreground stalks, Background field

STEP 3: THERMAL LAYER COMPOSITING



Insert Segmented Animal Between Occluding Layers Match Thermal Perspective

STEP 4: AUTO-LABELED THERMAL DATASET OUTPUT

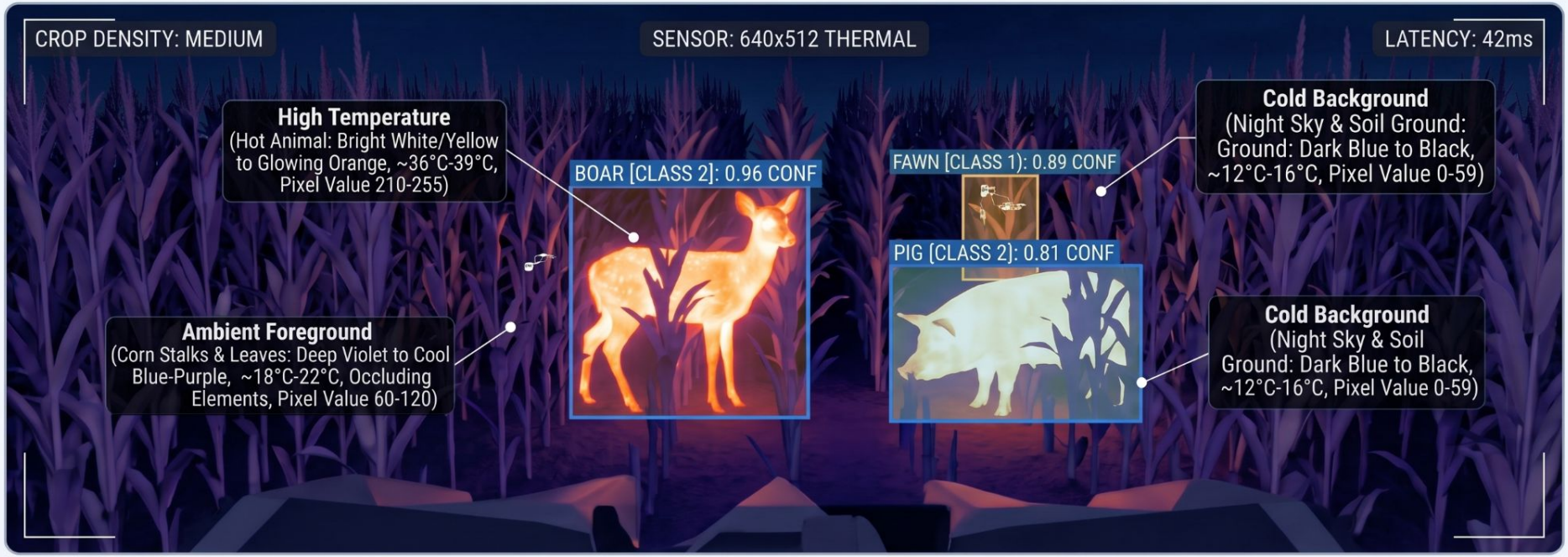


Diverse Thermal Data + Auto-Labels



YOLO Training Image + Automatic Label Export

THERMAL SIGNATURE & PIXEL RANGE BREAKDOWN (IRONBOW PALETTE)



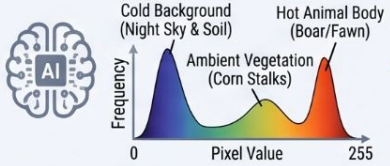
TECHNICAL PROCESS FLOW

STEP 1: THERMAL PALETTE ANALYSIS



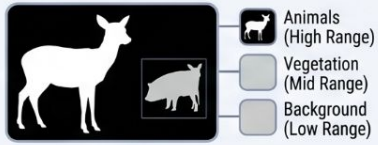
Color Palette Bar (Ironbow gradient with temperature scale from 12°C to 39°C and pixel intensity values 0 to 255)

STEP 2: PIXEL INTENSITY HISTOGRAM



Histogram / Pixel Value Distribution Curve

STEP 3: MULTI-THRESHOLD SEGMENTATION



Multi-Threshold / Semantic Segmentation Masks

STEP 4: THERMAL FEATURE EXTRACTION



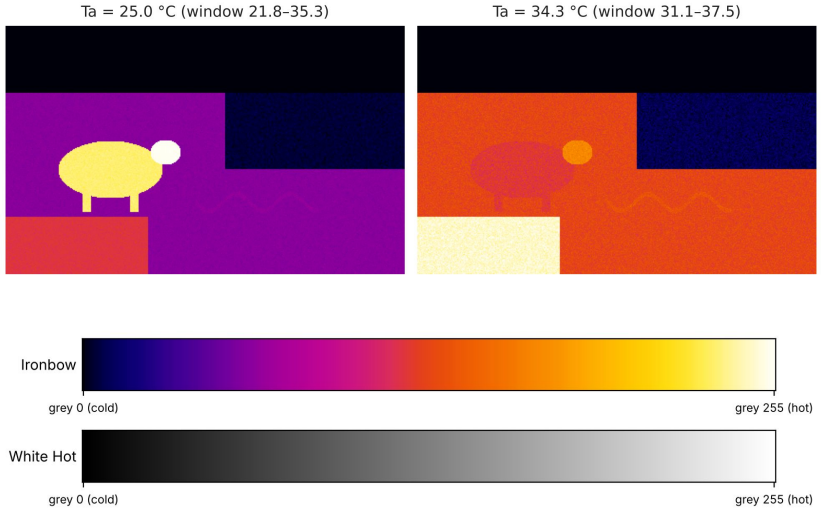
FEATURE VECTORS

Extracted Thermal Signatures (e.g., temperature profile, shape parameters) for detection and YOLO bounding box annotation

Synthetic Thermal Dataset in Blender

Segmenting animals hidden in crops during harvesting

Gujarat / West India · Air temperature (Ta) from a live Ahmedabad weather API



Illustrative schematic, not Blender output

What this deck covers

Thermal camera
How it works, FLIR in 8 steps

FLIR vs Blender
What we simulate and what we skip

Ta and window
Basics, and how we choose the window

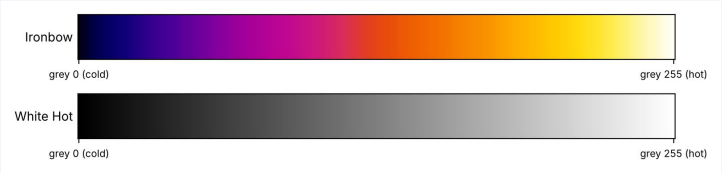
Class temperatures
Each class, with citations

Degrees to pixels
A small worked example

Evidence labels used throughout: SOURCED (cited measurement), PROXY (cited, related animal), ASSUMPTION (our choice).

What a thermal camera measures

- Every object emits infrared radiation set by its temperature and emissivity [1]
- Near-ambient objects radiate mostly in long-wave infrared (LWIR) [1]
- An uncooled microbolometer pixel is a tiny resistor that changes resistance as it warms [1]
- Glass blocks LWIR, so lenses are germanium [1]
- Colours are false colour from a palette; the palette changes the look, not the data [1][2]



Same grey values, two different palettes. Ironbow table: community repository [14].

How a FLIR camera works: 8 steps

1 Emission

Scene emits LWIR

[1]

2 Optics

Germanium lens focuses it

[1]

3 Detection

Microbolometer pixels change resistance

[1]

4 Digitise

ADC stores a raw 14/16-bit image

[1][3][4]

5 Correction

Pixel differences corrected (shutter NUC)

[1]

6 Radiometry

Raw counts become °C with calibration constants

[5]

7 AGC

Wide range squeezed to 8 bit for display

[3][6][7]

8 Palette

Grey mapped to White Hot, Ironbow, …

[1][2]

The eight-step split is our organisation of sources [1]–[7], not an official FLIR list. Not every FLIR core is radiometric [4].

FLIR vs our Blender pipeline

Stage	Real FLIR camera	Our Blender dataset
Scene temperature	From the real world	We assign °C to each class
Radiation, lens, detector	Physical (steps 1–3)	Not simulated
Raw counts	14/16-bit ADC output	Skipped: we store a °C map (32-bit float EXR)
Radiometric formula	Planck-type conversion to °C	Not needed: we start in °C
Noise and blur	Real sensor and lens	Added in code (amounts are an assumption)
Window (AGC)	Camera picks from the scene	**Per-image percentile stretch in code**
Palette	Camera lookup table	White Hot or Ironbow, HSV via OpenCV [8][9]
Segmentation mask	Manual labelling	**Free from Blender object IDs**

Synthetic infrared literature models temperature → radiance → grey level [10].

Some Terminologies

Ta (air temperature)

Live from the weather API for Ahmedabad. Fetched 14 Oct 2026: 34.3 °C, sunny.

Surface temperature

What the camera reads from skin, leaf or soil. Calf eye 35.3, coat 34.1 vs rectal 39.4 ° C [11].

Ectotherm vs endotherm

Snakes follow their surroundings; mammals keep themselves warm.

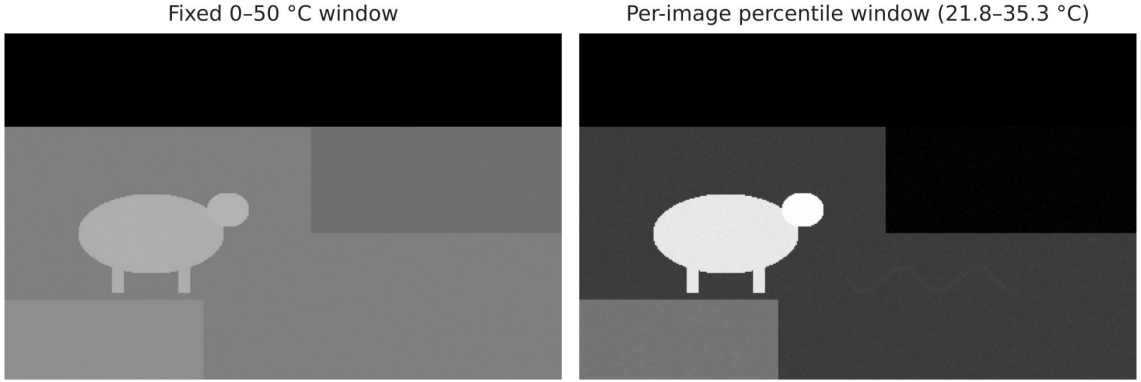
Thermal crossover

Animal and background at the same temperature: contrast vanishes [11].

Contrast

Animal minus background temperature. The window only decides how it is shown.

The window decides contrast

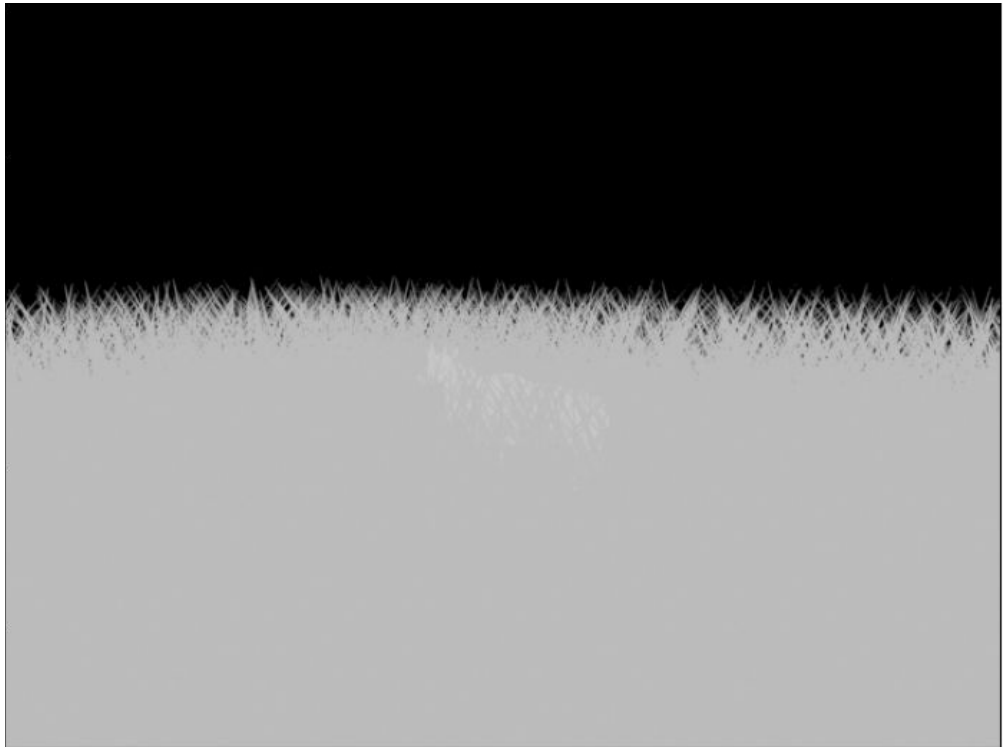


- Window = temperature range stretched to grey 0–255
- Narrow window: high contrast. Wide window: flat image
- Our earlier fixed 0–50 °C window was an arbitrary choice and made images flat

Illustrative schematic (Ta = 25 °C), not Blender output.

Same temperatures, very different pixels. Contrast depends on the window, so choose it with a documented rule.

Our earlier fixed 0–50 °C window



Our earlier fixed 0–50 °C window

How cameras choose, and what we do

Real cameras

- AGC re-maps the wide raw range to 8 bit, constantly adjusting to the scene [3]
- A new object can make the whole image brighter or darker [3]
- FLIR offers several AGC modes (linear, plateau, information-based) [6][7]
- A published baseline uses the 0.5 and 99.5 percentiles of the frame [7]

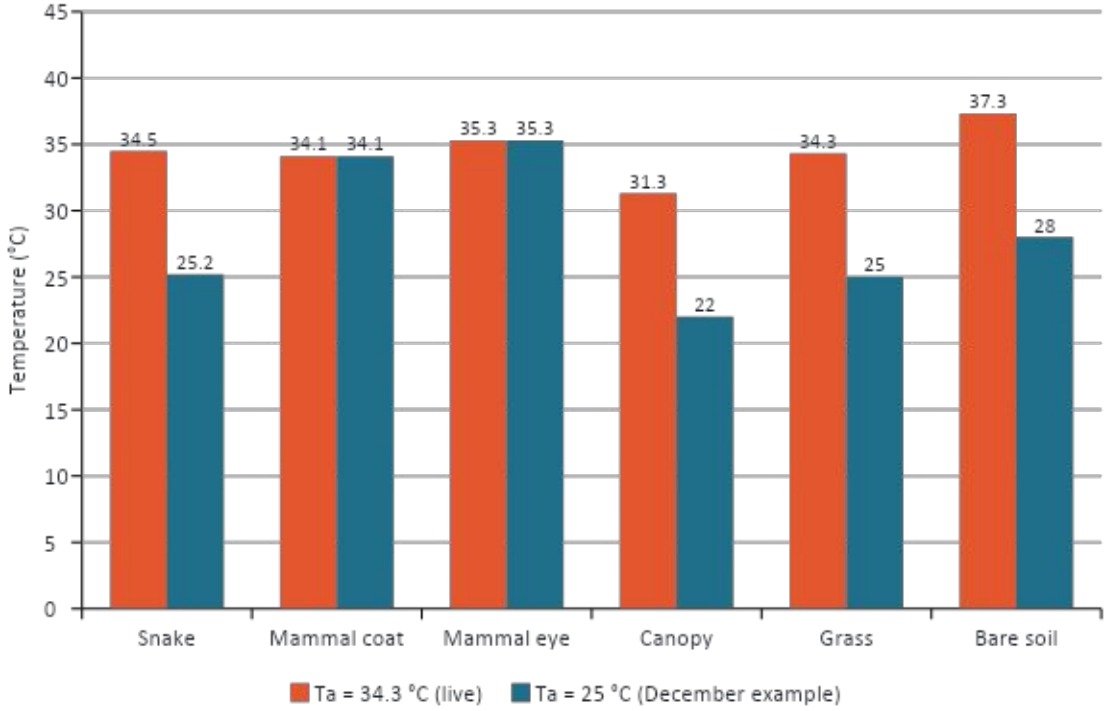
Our dataset

```
lo = percentile(T[no sky], 0.5)
hi = percentile(T[no sky], 99.5)
grey = clip(round(255*(T-lo)/(hi-lo)), 0, 255)
```

- Exclude the −40 °C sky from the percentiles, or it squashes everything else
- Keep the original °C EXR so another window can be applied later

The 0.5 / 99.5 values are a published baseline, not a law. Re-check [7] before submission.

Temperature of each class



Snake: Ta + 0.19 °C

SOURCED [12]

Mammal: absolute value

PROXY [11]

Canopy, grass, soil: Ta offsets

ASSUMPTION

Sky is fixed at −40 °C (assumption) and left off the chart.

Evidence behind each value

Class	Rule	Evidence	Detail and source
Snake	Ta + 0.19 °C	SOURCED	101 thermograms of 17 rattlesnakes: 0.19 °C above air (±0.11 SEM); US species [12]
Mammal (coat, eye)	34.1 / 35.3 °C absolute	PROXY	Dairy calves in hot weather, ambient 25.9–40.2 °C; eye 35.3, scapula 34.1, rectal 39.4 [11]
Bare soil	Ta + 3 °C	ASSUMPTION	USDA scene: soil 30.2 vs crops 19.2–26.3 °C at about 15:00, no air value [13]
Canopy	Ta − 3 °C	ASSUMPTION	Cooler than air is plausible; not verified this session
Grass	Ta + 0 °C	ASSUMPTION	No sourced offset
Sky	−40 °C, clipped	ASSUMPTION	No sourced value

No paper reports species-specific thermal pixel values for Gujarat animals or crops. Only the snake offset is directly sourced.

Degrees to pixels: a worked example

(a) Raw counts to °C (radiometric cameras) [5]

```
T = PB / ln( PR1 / (PR2*(raw+PO)) + PF ) - 273.15
```

raw count	temperature
18109	23.62 °C
18500	25.72 °C
19000	28.33 °C

Constants (PR1 21106.77, PB 1501, PF 1, PO −7340, PR2 0.012545258) are Thermimage's example from one camera.

(b) °C to grey (frame 22–34.1 °C)

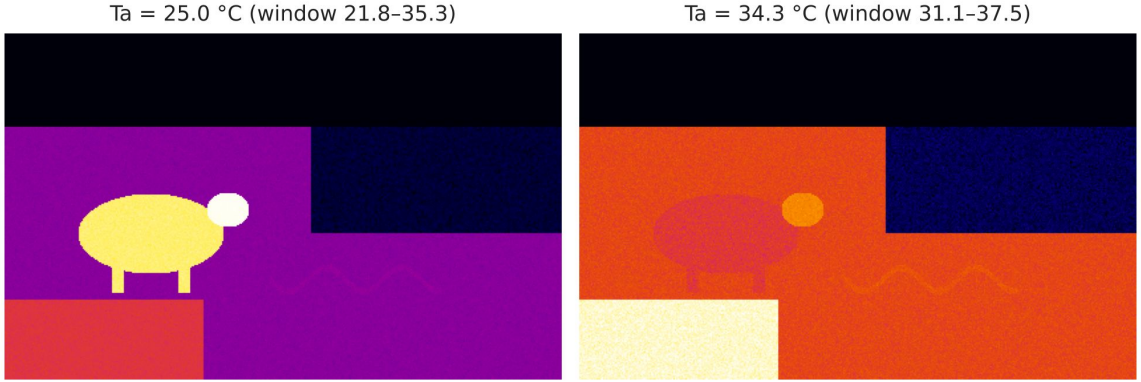
```
grey = 255 × (T − 22) / (34.1 − 22)
```

Object	T (°C)	Grey
Canopy	22.0	0
Grass	25.0	63
Snake	25.2	67
Calf coat	34.1	**255**

Fixed 0–50 °C window instead: calf only grey 174, grass 128.

(c) Grey → colour: White Hot uses R = G = B = grey; Ironbow looks up a 433-entry RGB table [14]; HSV via OpenCV [8].

Crossover: when Ta reaches animal temperature



- At 25 °C the calf is the hottest object
- At 34.3 °C the calf coat (grey 223) is darker than grass (239)
- Hot soil can outshine the animal

Illustrative schematic (Ironbow, per-image window), not Blender output.

A midday live Ta of about 34 °C removes most contrast. A model must not assume “hot means animal”.

Extra points and limits

Choose the time of day

Midday Ta (about 34 °C) kills contrast; December mornings and nights keep it. State the rule to maam.

Vary Ta and objects

Use several Ta values and ± a few °C per object (amounts are an assumption).

Save four outputs

Float °C EXR, 8-bit grey, palette image, and the object-ID mask.

Know the limits

Mammal value is from calves; canopy, grass, soil, sky are assumptions; Ironbow table is community-made [14].

References (Thermal Dataset Section)

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[10] Survey on Synthetic Infrared Image Synthesis. arXiv 2408.06868

[11] Bakony et al. Body surface temperatures of hutch-reared dairy calves. Frontiers in Veterinary Science, 2023

[12] Borrell LaDuc & Dudley. Rattlesnake thermography, 2005. repository.si.edu

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[14] FLIR-style thermal colour palettes, IRONBOW.c (community repository). github.com/MickTheMechanic



Implementation & Result

Frontend

COMPLETE

React Native

Backend

COMPLETE

Fast API

Database

NOT IMPLEMENTED

Database not implemented

ML Model

IN PROGRESS

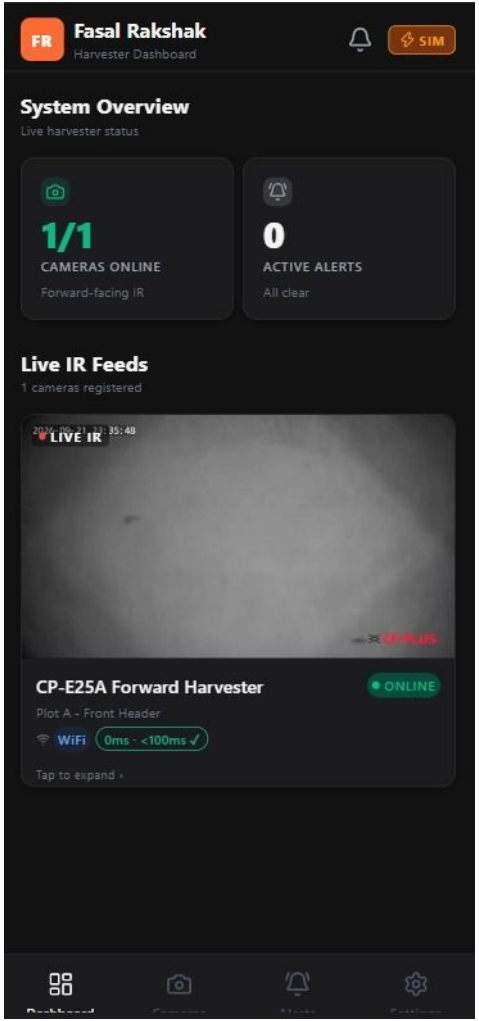
ML model in progress

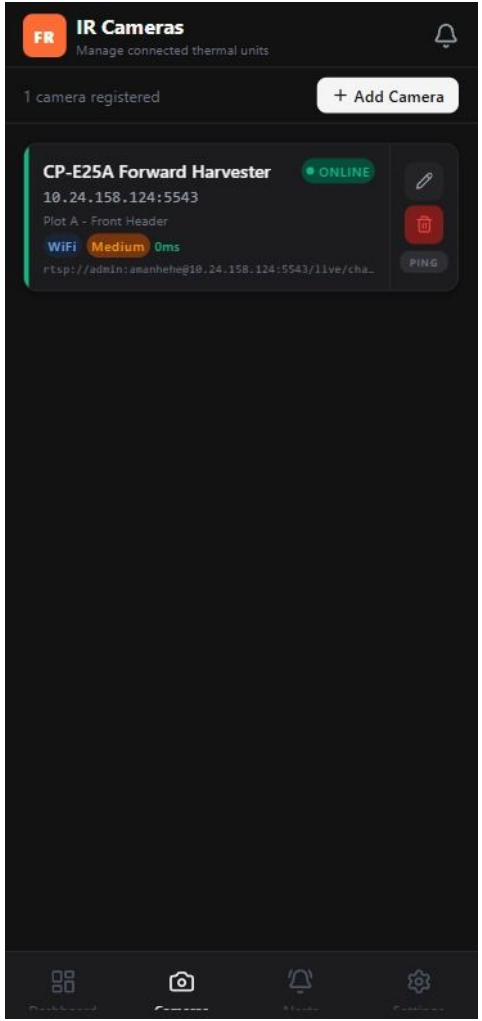
Baseline Vision Engine: Integrated a lightweight, single-stage deep learning model (YOLOv8) to handle real-time object detection.

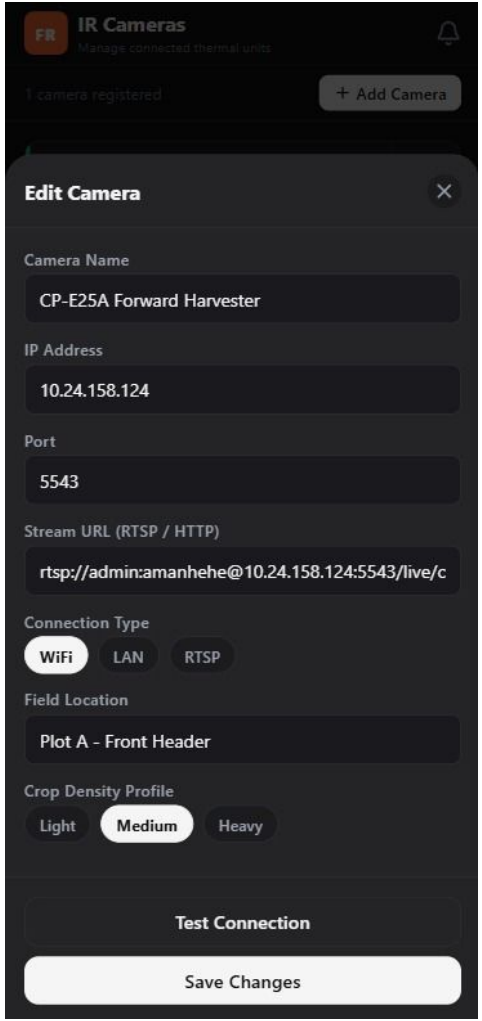
Live Pipeline Integration: Connected the vision model directly to a custom Python interface to capture live camera feeds and render bounding box overlays on screen.

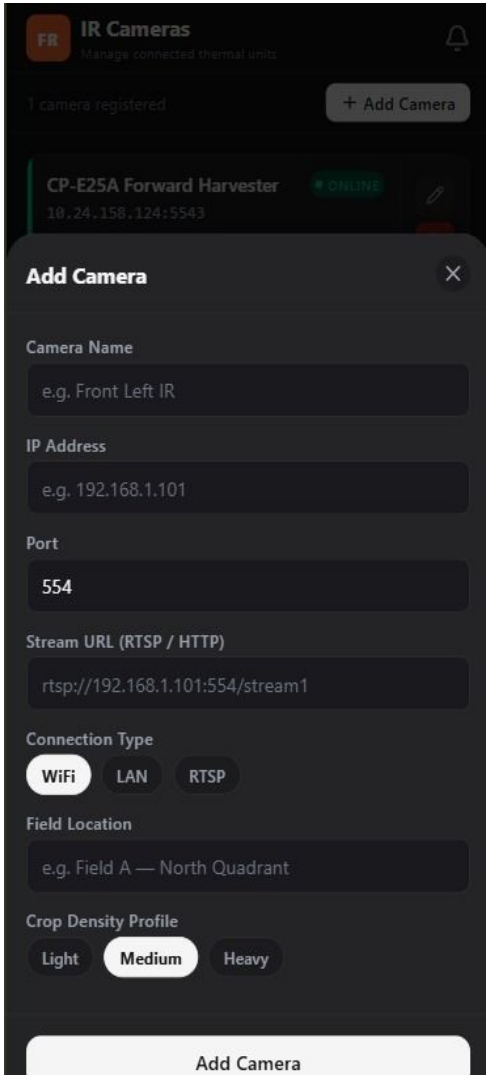
System Verification: Successfully tested real-time detection on everyday objects (mobile phone, mouse, remote) to confirm low latency and software stability.

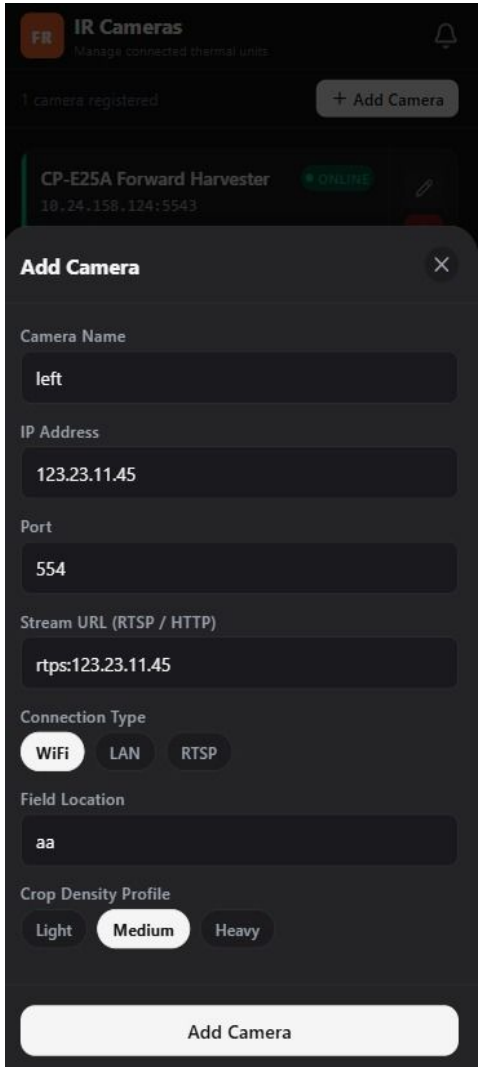
Modular Software Design: Built a flexible architecture so the baseline model can be easily updated or fine-tuned with custom field datasets without changing the main application.

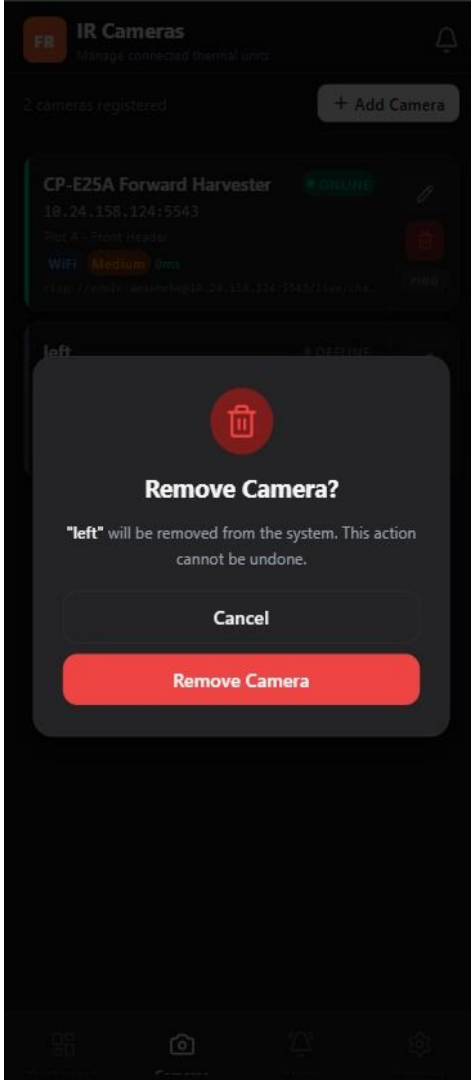

FR
Fasal Rakshak
Harvester Dashboard
SIM
System Overview
Live harvester status
1/1
CAMERAS ONLINE
Forward-facing IR
0
ACTIVE ALERTS
All clear
Live IR Feeds
1 cameras registered
LIVE IR
CP-E25A Forward Harvester
ONLINE
Plot A - Front Header
WiFi
0ms · <100ms ✓
Tap to expand ›

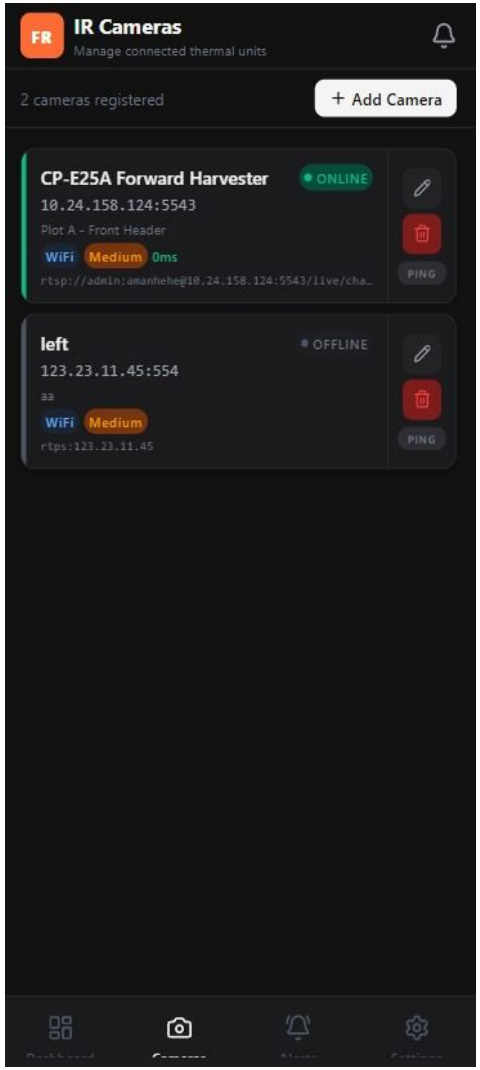

FR
IR Cameras
Manage connected thermal units
1 camera registered
+ Add Camera
CP-E25A Forward Harvester
ONLINE
10.24.158.124:5543
Plot A - Front Header
WiFi
Medium
0ms
PING


FR
IR Cameras
Manage connected thermal units
1 camera registered
+ Add Camera
Edit Camera
Camera Name
CP-E25A Forward Harvester
IP Address
10.24.158.124
Port
5543
Stream URL (RTSP / HTTP)
rtsp://admin:amanhehe@10.24.158.124:5543/live/c
Connection Type
WiFi
LAN
RTSP
Field Location
Plot A - Front Header
Crop Density Profile
Light
Medium
Heavy
Test Connection
Save Changes


IR Cameras
Manage connected thermal units
1 camera registered
+ Add Camera
CP-E25A Forward Harvester
ONLINE
10.24.158.124:5543
Add Camera
Camera Name
e.g. Front Left IR
IP Address
e.g. 192.168.1.101
Port
554
Stream URL (RTSP / HTTP)
rtsp://192.168.1.101:554/stream1
Connection Type
WiFi
LAN
RTSP
Field Location
e.g. Field A — North Quadrant
Crop Density Profile
Light
Medium
Heavy
Add Camera


IR Cameras
Manage connected thermal units
1 camera registered
+ Add Camera
CP-E25A Forward Harvester
ONLINE
10.24.158.124:5543
Add Camera
Camera Name
left
IP Address
123.23.11.45
Port
554
Stream URL (RTSP / HTTP)
rtps:123.23.11.45
Connection Type
WiFi
LAN
RTSP
Field Location
aa
Crop Density Profile
Light
Medium
Heavy
Add Camera


IR Cameras
Manage connected thermal units
2 cameras registered
+ Add Camera
CP-E25A Forward Harvester
ONLINE
10.24.158.124:5543
Plot A - Front Header
WiFi
Medium
0ms
PING
left
OFFLINE
Remove Camera?
"left" will be removed from the system. This action cannot be undone.
Cancel
Remove Camera


FR
IR Cameras
Manage connected thermal units
2 cameras registered
+ Add Camera
CP-E25A Forward Harvester
ONLINE
10.24.158.124:5543
Plot A - Front Header
WiFi
Medium
0ms
rtsp://admin:amanhehe@10.24.158.124:5543/live/cha...
PING
left
OFFLINE
123.23.11.45:554
aa
WiFi
Medium
rtps:123.23.11.45
PING

Fasal Rakshak

WILDLIFE DETECTED • ACTIVE

Emergency alert · 11:45:37 pm



CP-E25A Forward Harvester · Annotated snapshot

Mouse

TARGET SPECIES

CONFIDENCE 82.6%

SIoU Bounding Box · YOLO Edge Model

CROP OCCLUSION: Medium

SEVERITY: CRITICAL

CAMERA SOURCE: CP-E25A Forward Harvester

FIELD LOCATION: See camera details

⚠ Acknowledge / Slow Down

⊗ Dismiss — False Alarm

Fasal Rakshak

WILDLIFE DETECTED • ACTIVE

Emergency alert · 11:48:00 pm



CP-E25A Forward Harvester · Annotated snapshot

Mobile Phone

TARGET SPECIES

CONFIDENCE 94.8%

SIoU Bounding Box · YOLO Edge Model

CROP OCCLUSION: Medium

SEVERITY: CRITICAL

CAMERA SOURCE: CP-E25A Forward Harvester

FIELD LOCATION: See camera details

⚠ Acknowledge / Slow Down

⊗ Dismiss — False Alarm

FR **Alert History**

Detection audit log

41

Filters 59 events

Mobile Phone Just now
CP-E25A Forward Harvester
ACK Medium crop 77.1%

Mobile Phone Just now
CP-E25A Forward Harvester
ACTIVE Medium crop 70.4%

Mobile Phone Just now
CP-E25A Forward Harvester
ACTIVE Medium crop 75.3%

Mobile Phone Just now
CP-E25A Forward Harvester
ACTIVE Medium crop 92.4%

Mouse Just now
CP-E25A Forward Harvester
ACTIVE Medium crop 93.6%

Mobile Phone Just now
CP-E25A Forward Harvester
ACTIVE Medium crop 94.8%

Mouse Just now
CP-E25A Forward Harvester

9+ 9+



Camera Hardware & Integration

Department of Computer Science

Camera Hardware & Integration

CP-E25A Specifications

Sensor Type	2MP CMOS
Resolution / FPS	1080p 20 fps
IR Wavelength	850 nm IR
IR Range	About 10 m
Video Encoding	H.264 dual stream
Camera Type	Active-NIR prototyping
Thermal Capability	Not thermal



Future Work

FUTURE

Future Work Roadmap

1. Stream Ingestion
Fix stream ingestion

2. Database
Add the database

3. Model Training
Finish annotation & training

4. Benchmarks
Run ablation & edge benchmarks

5. Thermal Sensor
Select a thermal sensor

Expected Contribution

- An edge-deployable, occlusion-aware system
- A density-based limits characterization



THANK YOU

Proposed Edge-Deployable Wildlife Detection System

System Architecture — End-to-End Edge Pipeline for Real-Time Agricultural Wildlife Protection

1. ENVIRONMENT

Field Setup:

- Agricultural Field
- Tall crops
- Hidden animal
- Harvester

2. CAMERA

Forward Sensor:

IR / NIR Sensing

Current: CP-E25A (850nm IR)

Note: Thermal/IR sensor subject to ongoing hardware validation

3. STREAM

RTSP / Video:

Live Video Stream

Continuous real-time feed

4. PREPROCESS

Frame Prep:

- Frame Extraction
- Resize & Scale
- Normalization
- Noise / Clutter Filtering

5. MODIFIED YOLO DETECTOR

Coordinate Attention

Embeds positional location info into channel attention for enhanced spatial feature encoding.

Feature-Preserving Neck

High-Resolution multi-scale fusion to retain fine details of small/occluded targets.

SIoU Loss Function

Considers vector angle between mismatch to accelerate convergence & box precision.

6. OUTPUT

Detection Result:

- Animal Bounding Box Coordinates
- Confidence Score (%)
- Class Label

7. ALERT

In-Cab Alarm:

- Visual Flashing Alert
- Audible Warning Signal

Immediate hazard notification

8. OPERATOR

Human Receiver:

- Farmer / Harvester Driver

Receives alert to take immediate braking / avoidance action

EDGE COMPUTATION LAYER — LOCAL ON-DEVICE PROCESSING

Hardware: Embedded GPU / Edge Computing Unit mounted directly on Harvester vehicle.
Key Constraint: **NO CLOUD DEPENDENCY DURING INFERENCE** — 100% local processing guarantees zero latency loss & operational reliability in remote farm fields without cellular network coverage.

References (Intro & Animal Classes)

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[3] Cukor J. et al. (2025). No effect of a commercially used odor repellent for roe deer protection during meadow harvest. Animals 15(19):2932. mdpi.com/2076-2615/15/19/2932

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